MIL-A-46330(Ord)
15 February 1962
SUPERSEDING
FCS-822A
9 April 1958

MILITARY SPECIFICATION

AMPLIFIER, VIDEO: AM-1075 A/M

1. SCOPE

1.1 This specification covers one type of plan positioning indicator (PPI) video amplifier, hereinafter referred to as "video amplifier". It is a component of the assemblies and systems listed in 6.1.1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection Of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging Of
MIL-I-45208	Inspection Requirements: General Specification For

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1285

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DRAWINGS

Ordnance Corps

D7616975	Amplifier, Video: AM-1075 A/M Schematic Diagram
D8607305	Amplifier, Video: AM-1075 A/M

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one video amplifier, representative of production, shall be made available to the Government to determine if the video amplifier meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Design and construction.- The video amplifier shall be manufactured in accordance with Drawing D8607305 "Amplifier, Video: AM-1075 A/M" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence capabilities prior to final acceptance by the Government.

3.4 Ground potential.- The design and construction of the video amplifier shall insure that all external parts exclusive of electrical components, are at ground potential at all times.

3.5 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electron tubes
(f) Workmanship

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3.6 Environmental.

3.6.1 Operating temperatures. - The video amplifier shall meet all the performance requirements of this specification when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.6.2 Storage temperatures. - The video amplifier shall show no evidence of physical failure and shall meet all performance requirements of this specification at the standard ambient temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.6.3 Vibration. - The video amplifier shall be subjected to a simple harmonic motion having an amplitude of 0.015inch (0.03 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second (cps) return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration, the video amplifier shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.7 Electrical.

3.7.1 Circuitry. - The circuitry of the video amplifier shall be in accordance with the electrical schematic, Drawing D7616975.

3.7.2 Dielectric strength. - There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the video amplifier are subjected to an alternating current (ac) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 (cps) for a period of not more than 30 seconds nor less than 5 seconds.

3.7.3 Insulation resistance. - Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

3.8 Performance. - When set up in accordance with pretest information specified in Section 4, the video amplifier shall meet the following requirements (all symbols referred to are those shown on Drawing D7616975).

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3.8.1 Gate pulse output.

3.8.1.1 The output pulse at J2 shall have a pulse length adjustable over a minimum range of 735 plus or minus 25 microseconds (u sec) by means of Range Control R8, when a preknock pulse is applied to J1, and S1 on the 120,000 yard position.

3.8.1.2 The output pulse shall be 2.4 volts (min) with R8 set for a pulse length of 735 u sec.

3.8.1.3 The dc voltage at J2 shall be 80 percent (max) of the amplitude of the pulse specified in 3.8.1.2.

3.8.1.4 The tilt on the output waveform shall not exceed 10 percent of the amplitude of the 735 u sec pulse.

3.8.1.5 The output pulse at J2 shall have a pulse length adjustable over a minimum range of 395 plus or minus 15 u sec by means of R8, when a preknock pulse is applied to J1, and S1 on the 60,000 yard position.

3.8.1.6 The output pulse shall be 2.4 volts (min), with R8 set for a pulse length of 395 u sec.

3.8.2 Blanking pulse amplitude and clamping.

3.8.2.1 The dc voltage at the plate of V2B shall be 60 plus or minus 10 volts, when plus 80 volts is applied to terminal 5 of P1 (less preknock pulse).

3.8.2.2 The pulses (for each position of S1) at the plate of V2B shall have amplitudes of 100 volts (min). The tops of the pulses shall be flat within 10 percent (preknock pulse applied).

3.8.2.3 The dc voltage at the plate of V2B shall be 80 minus 0.6 of the pulse amplitude plus or minus 10 percent of the pulse amplitude of the 60,000 yard pulse in 3.8.2.2, when S1 is on the 60,000 yard position.

3.8.3 Low frequency response and clamping.

3.8.3.1 The amplitude of the output pulse at the plate of V4 shall be 25 volts (minimum peak to peak), when a 1 volt (peak to peak), 1000 u sec positive square wave pulse is applied to J3. The tilt shall not exceed 10 percent.

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3.8.3.2 The difference in dc plate voltage of V4, when the square wave pulse 3.8.3.1 is applied to J3 and not applied shall be the dc voltage plus 0.5 times the pulse amplitude of the output pulse referenced in 3.8.3.1 plus or minus 5 percent.

3.8.4 High frequency response.

3.8.4.1 The output at the plate of V4 shall have a minimum amplitude of 25 volts, when a positive 1.0 volt, 3.0 u sec, pulse at 1000 pulses per second (pps) is applied to J3.

3.8.4.2 The amplitude of the pulse shall not decrease to less than 0.9 of the value of the output pulse in 3.8.4.1 when the input in 3.8.4.1 is reduced in pulse width from 3.0 u sec to 0.6 u sec.

3.9 Identification marking. - Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's quality assurance system. - The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.1.1 The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

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4.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification shall consist of: (a) surveillance of the operation to determine whether the practices, methods, and procedures of the written plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrections have been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Contractor's inspection system. - Contractor's inspection and Government verification of contractor's inspection shall be in accordance with Specification MIL-I-45208.

4.4 Production model. - Unless otherwise specified in the contract, one video amplifier shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production model failure. - Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.5 Lot formation. - Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

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4.6 Contractor's inspection for all components not covered by SQAP. - Inspection for these components shall be in accordance with the inspection procedures of the written inspection plan required by 4.1.4.2 and 4.3. The plan shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the video amplifier.

4.7 Material. - Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.8 Inspection equipment. - Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.8.1 Accuracy requirements. - Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8.1.1 The following specific requirements are applicable to the items below:

Item No.	Description
1.	Generator, pulse. preknock, to produce a 1 u sec positive pulse of 25 volt amplitude at 1000 pps.
2.	Oscilloscope: Power supply: 115 ± 10 volts, 50-1600 cps single phase, 210 watts. Input impedance: 300,000 ohms paralleled by 30 micro-microfarads (uuf) Nominal signal input: 0.1 to 100 volts peak. Output: Trigger pulse ± 25 volts, 4 u sec duration, occuring 300,800, or 2,000 times per second, rising from 10-90 percent of full amplitude within 1/2 u sec; sawtooth voltage plus 150, available through 100,000 ohms in series with 50,000 uuf.

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Item No.	Description
	Frequency range: 1 cycle to 3 megacycles per second. Response drop 0.3 decibel at each end point.
	Sensitivity: A signal of 0.1 volt producing standard defection with max. amplification.
	Sweep time: Continuously adjustable from 0.5 - 50,000 u sec per inch. Start-Stop, each sweep started by signal pulses independent of preceding.
	Rise time: 0.08 u sec.
3.	Voltmeter, vacuum tube: 0-3 and 0-100 volts, accuracy ± 2 percent.
4.	Resistor: 220 ohms, 10 percent, 1/2 watt.
5.	Generator, square wave:
	Frequency range: Continuously variable 6-100,000 cycles.
	Wave shape: Rise time less than 0.2 u sec with negligible overshoot or 5 percent at 75 peak volts output.
	Output voltage: Step attenuator giving 75, 50, 25, 15, 10, 5 peak volts fixed and 0-2.5 volts continuously variable.
	Output impedance: 20 ohms per volt.
6.	Generator, to generate a variable 0.6 u sec to 3 u sec, 1 volt positive pulse.
7.	Power supply, + 250 volts, 0.030 ampere.
8.	Power supply, + 150 volts, 0.020 ampere.
9.	Power supply, -250 volts, 0.002 ampere.
10.	Power supply, 6.3 volts, 400 cycles, 2 amperes.

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4.9 Tests.

4.9.1 Classification of defects. - The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.9.1.1 Disposition of rejected lots. - Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None defined

MAJOR	AQL 6.5%	Req't Par.	QA Par.	Test Method
101.	Operating temperatures	3.6.1	4.9.2.1	Standard Measuring Equipment (SME)
102.	Storage temperatures	3.6.2	4.9.2.2	SME
103.	Vibration	3.6.3	4.9.2.3	Visual-Tactile Device, Special Testing (DST)

MINOR: None defined

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None defined

MAJOR:	AQL 1.5%	Req't Par.	QA Par.	Test Method
101.	Completeness of assy	3.3 & Dwg D8607305		Visual
102.	Ground potential	3.4 & Dwg D7616975		Visual
103.	Circuitry	3.7.1	4.9.3.1	Standard Measuring Equipment (SME)

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TABLE II - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 1.5%	Req't Par.	QA Par.	Test Method
104. Dielectric strength	3.7.2	4.9.3.2	Standard Measuring Equipment (SME)
105. Insulation resistance	3.7.3	4.9.3.3	SME
106. Gate pulse output	3.8.1	4.9.4.1	SME
107. Blanking pulse amplitude and clamping	3.8.2	4.9.4.2	SME
108. Low frequency response and clamping	3.8.3	4.9.4.3	SME
109. High frequency response	3.8.4	4.9.4.4	SME
110. Identification marking	3.9	4.9.5	Visual

MINOR: None defined

4.9.2 Environmental. - The test equipment used in conducting temperature tests shall be such that the units under test shall not exceed fifty percent of the internal volume of the test chamber, and no portion shall be closer than six inches from any surface of the chamber.

4.9.2.1 Operating temperatures. - The units being tested shall be installed in the test chamber in a manner similar to its normal operating position. The test shall then be conducted in accordance with Procedure I of Specification MIL-F-13926, to the conditions specified in 3.6.1. Upon completion of the exposure specified in 3.6.1, the unit under test shall be inspected as depicted herein to determine conformance with the requirements specified.

4.9.2.2 Storage temperatures. - The video amplifiers to be tested shall be installed in the test chamber in a manner similar to its normal operating position. The test shall then be conducted in accordance with Procedure II of Specification MIL-F-13926, to the conditions specified in 3.6.2. Upon completion of the exposures specified in 3.6.2, the unit under test shall show no evidence of damage by a visual and tactile examination. The unit then shall be subjected to the tests specified in 4.9.2.1 to 4.9.3.4 inclusive, and shall meet the requirements specified.

4.9.2.3 Vibration. - The video amplifier shall be clamped on the vibration table in its normal mounting or operating position. The equipment shall be vibrated in each of the three (3) mutually perpendicular directions that are respectively parallel to the edges of the equipment at the acceleration, frequency and duration specified in 3.6.3. At the conclusion of the test, the video amplifier shall be visually inspected for physical damage, then subjected to the tests specified in 3.6.3.

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4.9.3 Electrical.

4.9.3.1 Circuitry. - A continuity check of the video amplifier circuits shall be made using standard measuring equipment to determine conformance with the requirements specified.

4.9.3.2 Dielectric strength. - The video amplifier shall be tested in accordance with Method 301 of Standard MIL-STD-202, and shall meet the requirements specified.

4.9.3.3 Insulation resistance. - Immediately after satisfactorily withstanding the test for dielectric strength, the video amplifier shall be given the insulation resistance test in accordance with Method 302, test condition B, of Standard MIL-STD-202, and shall meet the requirements specified.

4.9.4 Performance. - Tests conducted to determine conformance with the requirements will be accomplished utilizing the test equipment and components specified in 4.8.1.1 and its accuracies, connected as shown in figure 1, or as described herein.

4.9.4.1 Gate pulse output.

4.9.4.1.1 With the 220 ohm, 1/2 watt resistor item 4 connected from J2 to ground, the preknock pulse from item 1 applied to J1, and switch S1 set in the 120,000 yard clockwise position; R8 shall be capable of adjusting the position gate over the minimum range specified in 3.8.1.1 as observed on oscilloscope, item 2.

4.9.4.1.2 R8 shall be set for the pulse length specified in 3.8.1.2. The amplitude of the gate shall be 2.4 volts minimum.

4.9.4.1.3 The dc voltage at J2, measured by the voltmeter item 3, shall be 0.80 percent maximum of the amplitude of the observed waveform measured in 4.9.4.1.2.

4.9.4.1.4 With a pulse length of 735 u sec, the tilted portion on the output waveform shall not exceed 10 percent of the waveform amplitude measured in 4.9.4.1.2.

4.9.4.1.5 With the preknock pulse applied to J1, and with switch S1 set to the 60,000 yard position, the pulse length shall be within the range specified in 3.8.1.5 by varying R8.

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4.9.4.1.6 Set the R8 for the pulse length specified in 3.8.1.6. The amplitude of the gate shall be 2.4 volts minimum.

4.9.4.2 Blanking pulse amplitude and clamping.

4.9.4.2.1 A positive 80 volts shall be applied to terminal 5 of P1 and disconnect the preknock pulse from J1. The dc voltage measured at pin 6 of V2B shall be as specified in 3.8.2.1.

4.9.4.2.2 Reapply the preknock pulse to J1. Using the oscilloscope item 2, observe and record the pulse amplitude at pin 6 of V2B for each position of switch S1. The amplitudes and the tops of the pulses shall be as specified in 3.8.2.2.

4.9.4.2.3 Set switch S1 to the 60,000 yard position. Using the voltmeter item 3, measure the dc voltage at pin 6 of V2B to determine conformance with the requirements specified in 3.8.2.3 in conjunction with the pulse amplitude measured in 4.9.4.2.2.

4.9.4.3 Low frequency response and clamping.

4.9.4.3.1 Using square wave generator item 5, a 1 volt, 1000 positive square wave pulse shall be applied to J3. With the oscilloscope connected to pin 5 of V4 observe the output pulse amplitude to determine conformance with the requirements specified in 3.8.3.1.

4.9.4.3.2 Using the voltmeter item 3, the dc voltage at pin 5 of V4 shall be measured and recorded.

4.9.4.3.3 Remove the square wave input from J3 and again the dc voltage at pin 5 of V4 shall be measured. This voltage shall be the reading of 4.9.4.3.2 plus 0.5 times the pulse amplitude of 4.9.4.3.1 plus or minus 5 percent.

4.9.4.4 High frequency response.

4.9.4.4.1 With a u sec, 1 volt positive pulse at 1000 pps, from generator item 6, applied to J3, the output at pin 5 of V4, as viewed by the oscilloscope, shall meet the requirement specified in 3.8.4.1.

4.9.4.4.2 The pulse width of 4.9.4.4.1 shall be reduced from 3.0 to 0.6. The amplitude of the output pulse shall not decrease to less than 0.9 of the value obtained in 4.9.4.4.1.

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4.9.5 Identification marking. - Marking for identification shall be visually inspected for conformance to Standard MIL-STD-130.

4.9.6 Preparation for delivery. - The quality assurance provisions for the preservation, packaging and packing of the amplifier shall be in accordance with the prescribed specification in 5.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. - The video amplifier shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619 P2063. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The purpose of the video amplifier is to amplify the video and range marks from the video and mark mixer to a value large enough for intensity modulation of the cathode-ray tube (CRT). It also supplies a positive square wave to the CRT for unblanking and a positive range gate to the (PPI) sweep generator in the radar receiver.

6.1.1 Application. - The video amplifier is a component of the following assemblies and systems:

Assembly	System
Console, Tracking	Antiaircraft Fire Control System M33C and M33D
Director, Computer Group OA-865/MSA-7	Radar Course Directing Control AN/MSW-1 and AN/MSW-3

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

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6.3 Definitions.

6.3.1 Standard measuring equipment (SME). - Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as: Meters, optical comparators, etc.

6.3.2 Special testing device (DST). - Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.3.3 Tactile. - Skillful manipulation or sensing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligations whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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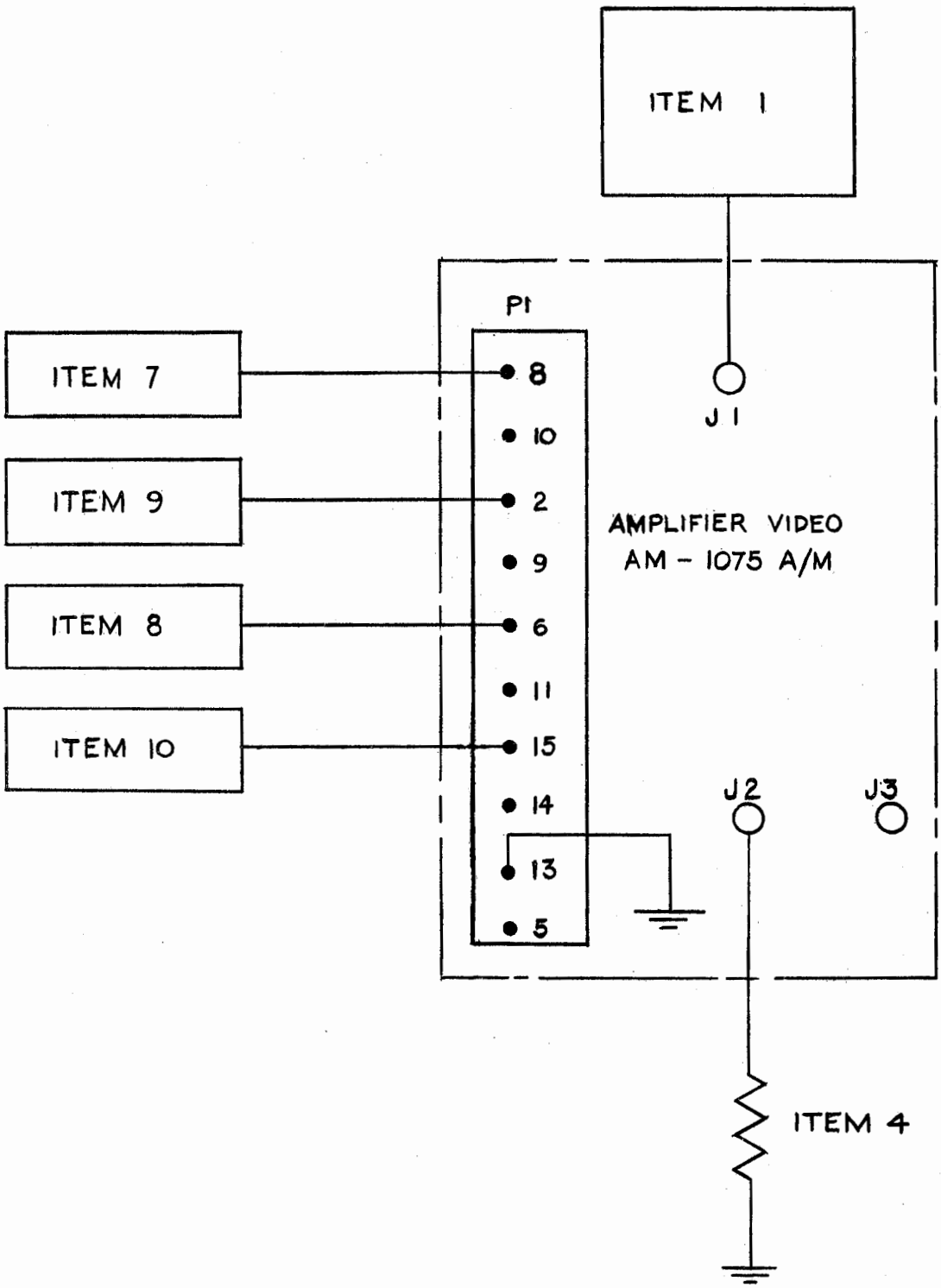


FIGURE 1

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